

Open-World Encrypted Network Anomaly Detection and Traffic Classification

NetAnomaly-OW | selective-state, attention, stabilized recurrent, and ensemble inference

Research question	Can encrypted applications be classified early while rejecting applications absent from training?
Evidence	Official CESNET-QUIC22 XS; 5,000 train, 1,000 calibration, 1,000 known-test and 1,000 held-out-unknown flows; chronological week-to-week split.

Abstract

This study investigates encrypted QUIC application recognition under an open-world setting, where test traffic may belong to services absent from training. The proposed system combines flow-level behavioral context, packet-sequence encoders, multi-prototype classification, conformal unknown rejection, early-flow evaluation, and chronological drift monitoring. The objective is not payload inspection or attack attribution, but reliable application classification and novelty triage from encrypted metadata.

1. Research Objective and Technical Contribution

Each bidirectional flow is represented by signed packet sizes, directions, inter-arrival times, aggregate statistics, and second-order path signatures. DriftMamba, a causal Transformer, and an xLSTM-inspired encoder share a hyperspherical multi-prototype head. Their heterogeneous ensemble combines complementary state-space, attention, and recurrent evidence. Split conformal calibration converts model evidence into known-traffic p-values and prediction sets; chronological windows provide drift auditing.

Packets + timing	Encoder	Context fusion	Prototype head	Conformal decision
up to 64 positions	Mamba / Transformer / xLSTM	aggregate + signatures	application similarity	KNOWN / UNKNOWN

2. Experimental Design and Leakage Controls

The chronological split is applied before feature fitting. Preprocessors, labels, anomaly thresholds, and conformal scores are learned only from training or calibration partitions. Testing uses the following week, preventing future-flow leakage and making unknown applications operationally meaningful.

Stage	Rows	Purpose	Temporal scope
Training	5,000	Learn encoders and prototypes	CESNET week 44
Calibration	1,000	Fit conformal rejection	CESNET week 44
Known test	1,000	15 trained applications	CESNET week 45
Unknown test	1,000	58 held-out applications	CESNET week 45

3. End-to-End Processing Pipeline

The system accepts official CESNET DataZoo frames or Wireshark-style packet exports. It constructs privacy-preserving bidirectional flows, derives synchronized sequence and statistical views, trains causal encoders, calibrates unknown rejection on a disjoint partition, and emits application, novelty, uncertainty, and drift outputs.

Stage	Input	Operation	Output
1. Ingestion	Packet/flow records	Schema validation and timestamp parsing	Canonical records
2. Flow building	Endpoints and protocol	Bidirectional grouping; endpoint redaction	Flow table
3. Representation	First packet positions	Sequence tensor, mask, aggregates, signatures	Multi-view tensors
4. Temporal split	Ordered flows	Train/calibration/known/unknown partitioning	Leakage-free subsets
5. Learning	Training subset	Encoder and prototype optimization	Neural checkpoint
6. Calibration	Known calibration subset	Split-conformal score fitting	P-value calibrator
7. Operation	New encrypted flow	Classify, reject, audit drift	Decision and evidence

4. End-to-End System Architecture

Figure 1 presents the complete learning and inference architecture. Encrypted packet or flow records are converted into privacy-preserving bidirectional flows and synchronized sequence, aggregate, and path-signature views. Three causal encoders provide complementary state-space, attention, and recurrent representations under an identical prototype and calibration protocol.

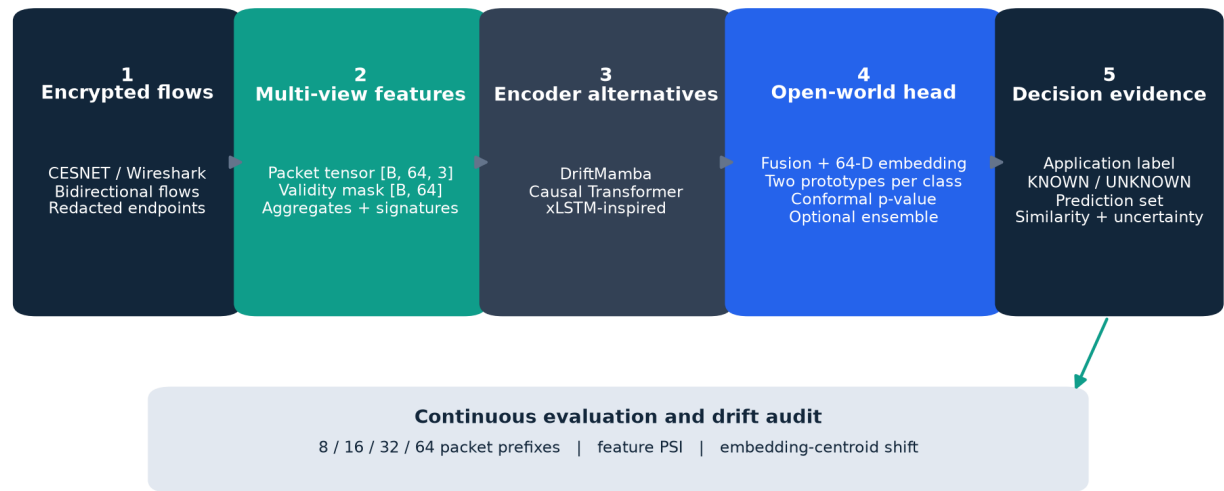


Figure 1. NetAnomaly-OW architecture from encrypted flow ingestion to calibrated operational decisions.

The encoders fuse masked sequence summaries with aggregate and signature projections before mapping each flow to a normalized 64-dimensional embedding. Two prototypes per known application model different traffic modes. Split-conformal calibration converts class evidence into prediction sets and knownness p-values; the optional heterogeneous ensemble aggregates the three application votes and calibrated p-values. Drift monitors operate alongside inference and do not update model weights on test traffic.

5. Multi-View Flow Representation

Packet payloads are never decrypted. The first packet positions are normalized into a causal tensor and accompanied by a mask so padded positions cannot influence pooling. Aggregate features provide global context, while second-order path signatures encode ordered cross-channel interactions. This combination follows the contextual-flow idea while remaining deployable from metadata.

Feature view	Representative variables	Purpose
Packet sequence	Signed size, direction, inter-arrival time, validity mask	Preserve order and request-response dynamics
Aggregate context	Duration, bytes, packet counts, ratios and distribution summaries	Describe complete flow behavior
Path signature	Second-order interactions over size, direction and time	Encode trajectory shape beyond simple moments
Operational context	Protocol/ports and chronological start time where available	Support routing, splitting and drift windows

6. Neural Pipelines and Model Roles

Pipeline	Sequence mechanism	Role and trade-off
DriftMamba	Input-selective causal state updates; linear scan	Strongest neural unknown separation; streaming-friendly
Causal Transformer	Masked multi-head self-attention with learned positions	Highest individual accuracy; quadratic attention cost
xLSTM-inspired	Stabilized exponential gates and normalized scalar memory	Strong early-packet behavior; weaker unknown separation
Deep ensemble	Majority label vote plus mean calibrated knownness	Best accuracy/macro-F1; extra inference cost

All three encoders project packet observations into a shared latent dimension, apply masked mean and last-valid-state pooling, fuse aggregate and signature projections, and normalize the final embedding. Each application owns multiple learnable prototypes so one service can express several traffic modes. Similarity to the nearest class prototype produces logits and a hard-negative margin encourages separation from confusing applications.

7. Detailed Neural Architecture

The shared input is a packet tensor of shape [batch, 64, 3] containing normalized signed size, direction and inter-arrival time, paired with a [batch, 64] validity mask. Packet observations are projected to 64 dimensions and processed by three causal blocks. Masked mean pooling and the last valid hidden state are fused with independently projected aggregate features and six-dimensional path-signature features. The fused representation is normalized to a 64-dimensional embedding; each known application is represented by two learnable prototypes to capture multiple behavioral modes.

Component	Implemented mechanism	Computational / modeling implication
DriftMamba	Input-dependent candidate, write, read and step gates; softplus-positive decay; masked recurrent state scan	Causal O(L) sequence processing and strongest unknown AUROC. It is Mamba-inspired pure PyTorch, not binary-equivalent to the official CUDA kernels.
Causal Transformer	Learned positions, four-head masked attention, key-padding mask, three pre-norm layers and GELU feed-forward blocks	Global packet interactions and highest individual accuracy, with $O(L^2)$ attention cost.
xLSTM-inspired	Stabilized exponential input/forget gates, running stabilizer and normalized scalar memory	Causal O(L) recurrence with good early-packet behavior; inspired by xLSTM rather than a full official-package reproduction.
Prototype head	Cosine similarity to two prototypes per class plus hard-negative separation	Supports multimodal application behavior and produces interpretable nearest-prototype evidence.

8. Optimization and Open-World Decision Logic

Weights are fitted only on the training partition; calibration data determine stopping and the empirical conformal score distribution, while the test partitions remain untouched. Prefix inference at 8, 16, 32 and 64 packets measures how quickly a reliable decision becomes possible. Population Stability Index over input features and movement of the embedding centroid provide complementary chronological drift signals.

Stage	Configuration	Rationale / emitted evidence
Optimization	AdamW; learning rate 1e-3; weight decay 1e-4; batch 128; gradient norm 1.0	Stable mini-batch fitting; the checkpoint is selected by calibration loss rather than test performance.
Objective	Cross-entropy + 0.25 x hard-negative prototype margin; margin 0.15	Combines closed-set discrimination with explicit separation from the nearest incorrect application.

Stage	Configuration	Rationale / emitted evidence
Conformal calibration	1,000 disjoint known flows; alpha = 0.10	Class p-values define the prediction set. A knownness p-value below 0.10 yields UNKNOWN.
Deep ensemble	Majority application vote; arithmetic mean of three calibrated knownness p-values	Combines complementary inductive biases; it is heterogeneous, not repeated seeds of one model.
Operational record	Label, p-value, prediction set/size, nearest-prototype similarity, embedding norm and decision	Preserves enough evidence for alert review, uncertainty analysis and drift monitoring.

Reproducibility artifacts include the neural checkpoint, aggregate preprocessor, conformal calibrator, per-flow prediction CSV and metrics JSON. Together these separate learned parameters, calibration state and evaluation evidence, allowing the same operating threshold to be audited or changed without retraining the encoder.

9. Comparative Model Performance

Accuracy measures total known-flow correctness; balanced accuracy weights every application equally; macro-F1 emphasizes minority classes; unknown AUROC evaluates threshold-independent separation between known and held-out applications. No single metric is sufficient for an open-world classifier.

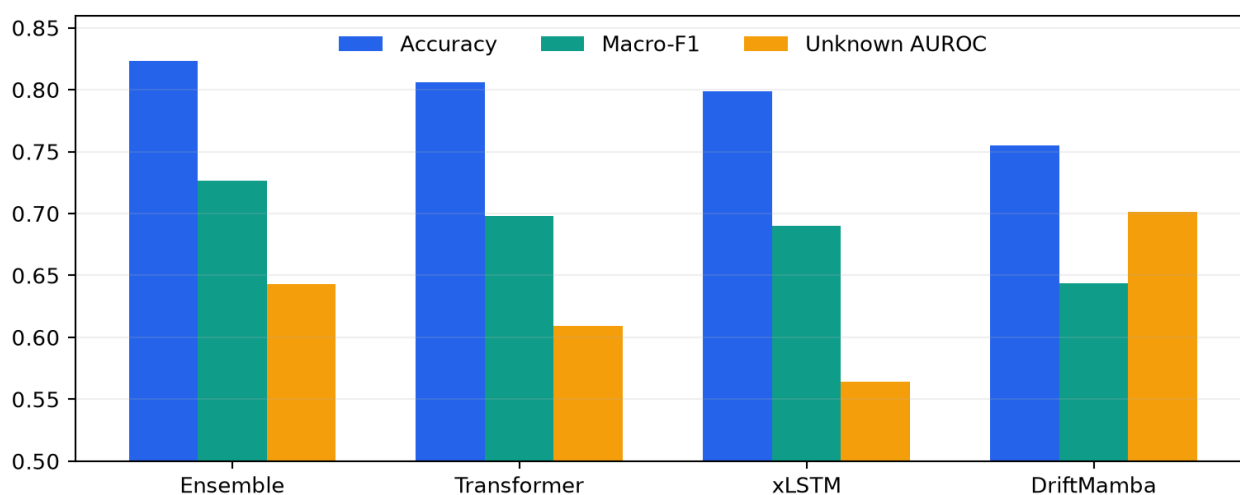


Figure 2. Final four-pipeline comparison on the same held-out flows.

Pipeline	Accuracy	Balanced acc.	Macro-F1	Unknown AUROC	Best use
Deep ensemble	0.823	0.7191	0.7264	0.6433	Overall classification
Transformer	0.806	0.7046	0.6981	0.6090	Raw accuracy
xLSTM-inspired	0.799	0.7087	0.6901	0.5643	Early classification
DriftMamba-12	0.755	0.6253	0.6435	0.7011	Unknown separation

The ensemble is heterogeneous: majority application voting across DriftMamba, Transformer and xLSTM, plus mean calibrated knownness. It is not three random seeds of one architecture. The ensemble maximizes classification quality, whereas DriftMamba provides the strongest neural unknown separation.

10. Open-World Flagging and Operational Risk

A flag means the calibrated known-traffic p-value is below 0.10. It is a novelty/review decision, not proof of an attack. Known flagged flows represent false rejection or rare legitimate behavior; unknown flagged flows represent successful detection of a held-out service. The table therefore separates alert volume from alert utility.

Pipeline	All flagged	Known flagged	Unknown flagged	Not flagged	Unknown recall	Known acceptance
DriftMamba	402	110	292	1,598	29.2%	89.0%
Transformer	278	88	190	1,722	19.0%	91.2%
xLSTM-inspired	237	92	145	1,763	14.5%	90.8%

Pipeline	All flagged	Known flagged	Unknown flagged	Not flagged	Unknown recall	Known acceptance
Deep ensemble	167	51	116	1,833	11.6%	94.9%

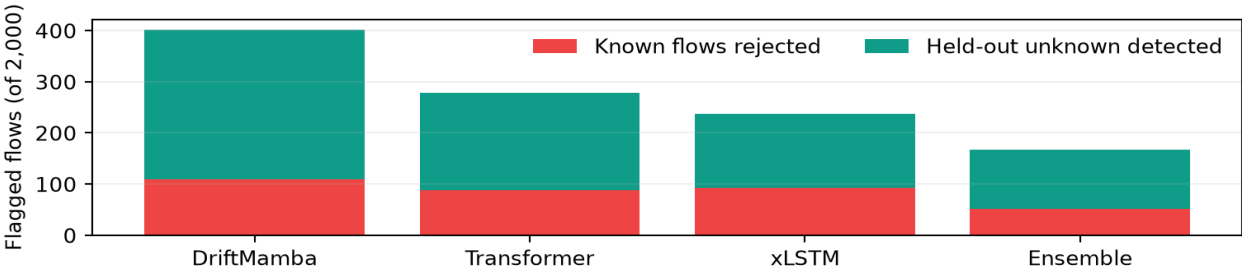


Figure 3. Composition of model flags. More flags are not automatically better.

11. Reproducibility and Repository Deliverables

The repository separates reusable source code, experiment configuration, validation, analysis notebooks, saved evidence and technical documentation. This structure supports transparent reruns, focused review and extension of each modelling pipeline.

Deliverable	Repository location	Purpose
Python package	src/driftmamba	Data, models, calibration and inference
Automated verification	tests/ and .github/workflows/ci.yml	Regression tests and lint checks
Reproducible analysis	notebooks/ and results/	Data plots and benchmark comparisons
Experiment automation	scripts/ and configs/	Dataset export, training and report generation
Technical evidence	output/pdf/ and docs/	Architecture, results, limitations and handoff

12. Evidence Gaps and Threats to Validity

The present results are genuine but bounded: they use one dataset edition, one chronological transition, and one random seed. The following evidence is required before making strong generalization or deployment claims.

Priority	Missing comparison parameter	Why it matters
High	3-5 random seeds with confidence intervals	Current values are single-seed estimates
High	Inference latency, throughput, memory and parameter count	Tests real-time feasibility
High	Per-class precision/recall/F1 and support	Exposes minority-service failures
High	Threshold curves: ROC, precision-recall and risk-coverage	Avoids one-threshold conclusions
Medium	External validation, ablations and ensemble agreement	Tests generalization and novelty claims

13. Conclusions and Recommended Operating Modes

The heterogeneous ensemble is the preferred classification mode, reaching 0.823 accuracy and 0.7264 macro-F1. DriftMamba-12 is the unknown-sensitive option, with 0.7011 unknown AUROC and 292 held-out detections at 89.0% known acceptance. The Transformer prioritizes raw accuracy, while xLSTM is useful at early packet budgets. The large not-flagged count reflects conservative rejection and should change only after measuring known-flow false rejection. The defensible contribution is a complementary four-pipeline comparison under a leakage-free open-world protocol, not production attack attribution.

NetAnomaly-OW | Results generated from saved experiment artifacts.